



IC 230: Environmental Science

Course Code	IC 230
Course Name	Environmental Science
L-T-P-C	3-0-0-3
Prerequisites	Consent of the faculty member
Intended for	B.Tech
Distribution	Core
Approval	33rd BoA (Jan 2020); 24th Senate (Feb 2020)

Course Contents

- **Environment components and issues:** Definition and Scope, Atmosphere, Hydrosphere, Lithosphere, Biosphere, Global, Regional and Local Issues, Major environmental issues faced by the world [12 Lectures]
- **Environmental systems:** Characteristics and properties, biogeochemical cycles of carbon, nitrogen, phosphorous and water, Biotic and abiotic environment, Food chain and webs, Anthropogenic influence on food chain [8 Lectures]
- **Natural Resources:** Forests, Energy resources, renewable and non-renewable energy, Mineral resources, Water resources [6 Lectures]
- **Environmental Pollution:** Common pollutants and their spread, Air pollution, Water pollution, Soil pollution, Measurement of pollution, Environmental parameters and standards, Environmental impacts and assessment [8 Lectures]
- **Environmental Management:** Monitoring and remediation, Reduction-reuse-recycling possibilities, Environmental policies, Case studies [8 Lectures]

Text Books

1. G. Tyler Miller and Scott Spoolman, **Environmental Science**, 16th edition, CENGAGE Learning Custom Publishing, Canada, 2017

References

1. Arvind Kumar, **A Text Book of Environmental Science**, P H Publishing Corporation, New Delhi, India.
2. Michael Alleby, **Basics of Environmental Science**, 2nd Edition, - Taylor & Francis Group, New York, USA, 2000